Linux Networking Commands

1. arp

Description: Displays and manages the system's ARP (Address Resolution Protocol) table, which maps IP addresses to MAC addresses on the local network.

Syntax

```
arp [options]
arp -a
arp -n
arp -d <IP>
```

Terminal

```
isha@Ubuntu:~/Desktop$ arp -a
_gateway (192.168.1.1) at 3c:84:6a:1b:2e:10 [ether] on eth0
printer.local (192.168.1.15) at 8c:ec:4b:aa:12:3d [ether] on eth0
laptop2.local (192.168.1.22) at f0:18:98:44:5c:9a [ether] on eth0
```

2. curl

Description: Transfers data to or from a server; commonly used to test URLs, download files, and interact with REST APIs.

Syntax

```
curl [options] URL
curl URL
curl -I URL
curl -v URL
curl -O URL
curl -X GET/POST URL
```

Terminal

```
isha@Ubuntu:~/Desktop$ curl -I https://example.com
HTTP/2 200
content-type: text/html; charset=UTF-8
date: Thu, 11 Sep 2026 09:14:02 GMT
content-length: 1256
server: ECS (dcb/7EEC)
```

3. host

Description: Performs DNS lookups to resolve domain names to IP addresses, or to query specific DNS record types.

Syntax

```
host [options] domain
host domain.com
host -t A domain.com
host -t MX domain.com
```

Terminal

```
isha@Ubuntu:~/Desktop$ host -t MX gmail.com
gmail.com mail is handled by 5 gmail-smtp-in.l.google.com.
gmail.com mail is handled by 10 alt1.gmail-smtp-in.l.google.com.
gmail.com mail is handled by 20 alt2.gmail-smtp-in.l.google.com.
```

4. hostid

Description: Displays the unique 32-bit hexadecimal identifier assigned to the host.

Syntax

```
hostid
```

Terminal

```
isha@Ubuntu:~/Desktop$ hostid
007f0101
```

5. hostname

Description: Displays or sets the system's hostname; can also show IP addresses associated with the host.

Syntax

```
hostname [options]
hostname
hostname -I
hostname -i
hostname -f
```

Terminal

```
isha@Ubuntu:~/Desktop$ hostname -I
192.168.1.42 172.17.0.1
```

6. hostnamectl

Description: Views or changes the system hostname and related settings, managed by systemd.

Syntax

```
hostnamectl [command]
hostnamectl
hostnamectl status
hostnamectl set-hostname NAME
```

Terminal

```
isha@Ubuntu:~/Desktop$ hostnamectl status
   Static hostname: dev-workstation
         Icon name: computer-laptop
           Chassis: laptop
        Machine ID: 8f3a9c2e1b4d4a7e9c6f0a1b2c3d4e5f
           Boot ID: 1a2b3c4d5e6f7081920a1b2c3d4e5f60
  Operating System: Ubuntu 24.04.1 LTS
            Kernel: Linux 6.8.0-45-generic
      Architecture: x86-64
```

7. ifconfig

Description: Legacy tool to view or configure network interfaces (IP address, netmask, status).

Syntax

```
ifconfig [interface] [options]
ifconfig
ifconfig eth0
ifconfig -a
ifconfig eth0 up/down
```

Terminal

```
isha@Ubuntu:~/Desktop$ ifconfig eth0
eth0: flags=4163<UP,BROADCAST,RUNNING,MULTICAST>  mtu 1500
        inet 192.168.1.42  netmask 255.255.255.0  broadcast 192.168.1.255
        inet6 fe80::215:5dff:fe00:12  prefixlen 64  scopeid 0x20<link>
```

```
        ether 00:15:5d:00:00:12  txqueuelen 1000  (Ethernet)
        RX packets 84213  bytes 98234112 (98.2 MB)
        TX packets 51204  bytes 14382011 (14.3 MB)
```

8. iftop

Description: Displays a live, continuously updating view of bandwidth usage on a network interface, per connection.

Syntax

```
iftop [options]
sudo iftop
iftop -i eth0
iftop -n
iftop -P
```

Terminal

```
isha@Ubuntu:~/Desktop$ sudo iftop -i eth0 -n
interface: eth0
IP address is: 192.168.1.42
                 2.5Kb  5.0Kb  7.5Kb  10.0Kb  12.5Kb
192.168.1.42  =>  142.250.193.14        1.20Kb  0.98Kb  1.05Kb
192.168.1.42  <=  142.250.193.14        4.30Kb  3.87Kb  4.01Kb
192.168.1.42  =>  17.253.144.10          320b   410b   380b
TX: cum:  1.2MB  peak:  8.4Kb  rates: 1.1Kb 1.0Kb 1.0Kb
RX: cum:  3.9MB  peak: 12.1Kb  rates: 4.3Kb 4.0Kb 4.0Kb
```

9. ip

Description: Modern replacement for ifconfig/route; manages IP addresses, links, routing tables, and neighbor (ARP) entries.

Syntax

```
ip [object] [command]
ip a
ip link
ip route
ip neigh
```

Terminal

```
isha@Ubuntu:~/Desktop$ ip a
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536
    inet 127.0.0.1/8 scope host lo
2: eth0: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500
    link/ether 00:15:5d:00:00:12 brd ff:ff:ff:ff:ff:ff
    inet 192.168.1.42/24 brd 192.168.1.255 scope global eth0
```

10. ipcrm

Description: Removes System V inter-process communication (IPC) resources such as shared memory, message queues, or semaphores.

Syntax

```
ipcrm [options] ID
ipcrm -m ID
ipcrm -q ID
ipcrm -s ID
```

Terminal

```
isha@Ubuntu:~/Desktop$ ipcrm -m 65536
isha@Ubuntu:~/Desktop$ echo $?
0
```

11. ipcs

Description: Displays information about active System V IPC resources: shared memory segments, message queues, and semaphores.

Syntax

```
ipcs [options]
ipcs
ipcs -m
ipcs -q
ipcs -s
```

Terminal

```
isha@Ubuntu:~/Desktop$ ipcs -m
------ Shared Memory Segments --------
key        shmid      owner      perms      bytes      nattch     status
0x00000000 65536      user       600        524288     2
```

12. iptables

Description: Configures the Linux kernel firewall — sets rules to allow, block, or forward network packets.

Syntax

```
iptables [options]
iptables -L
iptables -L -n -v
iptables -S
iptables -t nat -L
```

Terminal

```
isha@Ubuntu:~/Desktop$ sudo iptables -L -n -v
Chain INPUT (policy ACCEPT 1204 packets, 98K bytes)
pkts bytes target  prot opt in    out   source        destination
  50  3200 ACCEPT  tcp  --  eth0  *     0.0.0.0/0     0.0.0.0/0    tcp dpt:22
Chain FORWARD (policy DROP 0 packets, 0 bytes)
Chain OUTPUT (policy ACCEPT 980 packets, 76K bytes)
```

13. iptables-save

Description: Dumps the current iptables rule set to standard output in a format that can be restored later.

Syntax

```
iptables-save [options]
iptables-save
sudo iptables-save > rules.txt
```

Terminal

```
isha@Ubuntu:~/Desktop$ sudo iptables-save > rules.txt
isha@Ubuntu:~/Desktop$ cat rules.txt
# Generated by iptables-save v1.8.10 on Thu Sep 11 09:20:11 2026
*filter
:INPUT ACCEPT [1204:98213]
:FORWARD DROP [0:0]
:OUTPUT ACCEPT [980:76001]
-A INPUT -p tcp -m tcp --dport 22 -j ACCEPT
COMMIT
```

14. iwconfig

Description: Views or configures wireless network interface parameters such as SSID, mode, and signal strength.

Syntax

```
iwconfig [interface] [options]
iwconfig
iwconfig wlan0
iwconfig wlan0 essid NAME
```

Terminal

```
isha@Ubuntu:~/Desktop$ iwconfig wlan0
wlan0     IEEE 802.11  ESSID:"HomeNet-5G"
          Mode:Managed  Frequency:5.18 GHz  Access Point: 3C:84:6A:1B:2E:10
          Bit Rate=270 Mb/s   Tx-Power=20 dBm
          Link Quality=58/70  Signal level=-52 dBm
          Rx invalid nwid:0  Rx invalid crypt:0  Rx invalid frag:0
```

15. nc (netcat)

Description: Reads and writes raw data across network connections; used to test open ports, transfer data, or set up simple listeners.

Syntax

```
nc [options] host port
nc -zv HOST PORT
nc -l PORT
nc -u HOST PORT
```

Terminal

```
isha@Ubuntu:~/Desktop$ nc -zv 192.168.1.15 22
Connection to 192.168.1.15 22 port [tcp/ssh] succeeded!
```

16. netstat

Description: Displays active network connections, listening ports, and network statistics (legacy tool, superseded by ss).

Syntax

```
netstat [options]
netstat -a
netstat -t
netstat -u
netstat -l
netstat -tulpn
```

Terminal

```
isha@Ubuntu:~/Desktop$ netstat -tulpn
Active Internet connections (only servers)
Proto Recv-Q Send-Q Local Address     Foreign Address   State    PID/Program
tcp        0      0 0.0.0.0:22        0.0.0.0:*         LISTEN   812/sshd
tcp        0      0 127.0.0.1:3306    0.0.0.0:*         LISTEN   1420/mysqld
udp        0      0 0.0.0.0:68        0.0.0.0:*                  654/dhclient
```

17. nmcli

Description: Command-line interface for NetworkManager; manages network devices, connections, and Wi-Fi.

Syntax

```
nmcli [object] [command]
nmcli device status
```

```
nmcli connection show
nmcli device wifi list
```

Terminal

```
isha@Ubuntu:~/Desktop$ nmcli device status
DEVICE  TYPE      STATE      CONNECTION
eth0    ethernet  connected  Wired connection 1
wlan0   wifi      disconnected  --
lo      loopback  unmanaged  --
```

18. nslookup

Description: Queries DNS servers to resolve domain names to IP addresses or vice versa.

Syntax

```
nslookup domain [server]
nslookup domain.com
nslookup domain.com 8.8.8.8
```

Terminal

```
isha@Ubuntu:~/Desktop$ nslookup example.com 8.8.8.8
Server:  8.8.8.8
Address: 8.8.8.8#53

Non-authoritative answer:
Name:  example.com
Address: 93.184.215.14
```

19. ping

Description: Tests connectivity to a host by sending ICMP echo requests and measuring the response time.

Syntax

```
ping [options] host
ping HOST
ping -c 4 HOST
ping -4 HOST
ping -6 HOST
```

Terminal

```
isha@Ubuntu:~/Desktop$ ping -c 4 google.com
PING google.com (142.250.193.14) 56(84) bytes of data.
64 bytes from 142.250.193.14: icmp_seq=1 ttl=117 time=12.4 ms
64 bytes from 142.250.193.14: icmp_seq=2 ttl=117 time=11.9 ms
64 bytes from 142.250.193.14: icmp_seq=3 ttl=117 time=13.1 ms
64 bytes from 142.250.193.14: icmp_seq=4 ttl=117 time=12.0 ms

--- google.com ping statistics ---
4 packets transmitted, 4 received, 0% packet loss, time 3004ms
rtt min/avg/max/mdev = 11.9/12.35/13.1/0.44 ms
```

20. rcp

Description: Copies files between remote systems using the older, unencrypted 'r' protocol (largely replaced by scp/rsync).

Syntax

```
rcp [options] source destination
rcp file user@host:/path
rcp -r folder user@host:/path
```

Terminal

```
isha@Ubuntu:~/Desktop$ rcp notes.txt raj@192.168.1.20:/home/raj/
isha@Ubuntu:~/Desktop$ echo $?
0
```

21. route

Description: Displays or manually manipulates the IP routing table (legacy tool, superseded by ip route).

Syntax

```
route [options]
route -n
route
route add
route del
```

Terminal

```
isha@Ubuntu:~/Desktop$ route -n
Kernel IP routing table
Destination     Gateway         Genmask         Flags Metric Iface
0.0.0.0         192.168.1.1     0.0.0.0         UG    100    eth0
192.168.1.0     0.0.0.0         255.255.255.0   U     100    eth0
```

22. rsync

Description: Efficiently synchronizes or copies files and directories, locally or over a network, transferring only changed data.

Syntax

```
rsync [options] source destination
rsync -av
rsync -avn
rsync -e ssh
```

Terminal

```
isha@Ubuntu:~/Desktop$ rsync -av /home/user/docs/ backup@192.168.1.30:/backups/docs/
sending incremental file list
report.docx
notes.txt

sent 45,213 bytes  received 74 bytes  18,114.80 bytes/sec
total size is 44,980  speedup is 0.99
```

23. scp

Description: Securely copies files between hosts over an encrypted SSH connection.

Syntax

```
scp [options] source destination
scp file user@host:/path
scp -r
scp -P PORT
```

Terminal

```
isha@Ubuntu:~/Desktop$ scp report.pdf raj@192.168.1.20:/home/raj/docs/
raj@192.168.1.20's password:
report.pdf                100%  2.1MB   5.4MB/s   00:00
```

24. ssh

Description: Establishes a secure, encrypted remote login session or executes commands on a remote host.

Syntax

```
ssh [options] user@host
ssh user@host
ssh -p PORT
ssh -i KEY
ssh -v
```

Terminal

```
isha@Ubuntu:~/Desktop$ ssh raj@192.168.1.20
raj@192.168.1.20's password:
Welcome to Ubuntu 24.04.1 LTS (GNU/Linux 6.8.0-45-generic x86_64)
Last login: Wed Sep 10 21:12:44 2026 from 192.168.1.42
raj@server:~$
```

25. tracepath

Description: Traces the network path to a destination host and reports MTU along the route, without requiring root privileges.

Syntax

```
tracepath [options] host
tracepath domain.com
tracepath -n HOST
```

Terminal

```
isha@Ubuntu:~/Desktop$ tracepath example.com
1:  192.168.1.1                    0.412ms pmtu 1500
1:  192.168.1.1                    0.389ms
2:  10.10.0.1                     8.213ms
3:  no reply
4:  93.184.215.14                 12.882ms reached
    Resume: pmtu 1500 hops 4 back 4
```

26. traceroute

Description: Displays the route and measures transit delays of packets across a network to a destination host.

Syntax

```
traceroute [options] host
traceroute HOST
traceroute -n HOST
traceroute -I HOST
```

Terminal

```
isha@Ubuntu:~/Desktop$ traceroute -n example.com
traceroute to example.com (93.184.215.14), 30 hops max, 60 byte packets
1  192.168.1.1     0.512 ms  0.487 ms  0.465 ms
2  10.10.0.1       8.112 ms  8.045 ms  7.998 ms
3  100.65.0.1      9.877 ms  9.654 ms  9.532 ms
4  93.184.215.14  12.664 ms  12.590 ms  12.511 ms
```

27. vnstat

Description: Monitors and logs network traffic over time, providing daily, monthly, and historical usage statistics.

Syntax

```
vnstat [options]
vnstat
```

```
vnstat -i eth0
vnstat -d
vnstat -m
```

Terminal

```
isha@Ubuntu:~/Desktop$ vnstat -d
                     rx      |     tx      |    total    |   avg. rate
------------------------+-------------+-------------+---------------
2026-09-09    412.3 MiB | 88.1 MiB    |  500.4 MiB  |   46.36 kbit/s
2026-09-10    530.7 MiB | 102.4 MiB   |  633.1 MiB  |   58.62 kbit/s
2026-09-11    120.5 MiB | 30.2 MiB    |  150.7 MiB  |   65.11 kbit/s
------------------------+-------------+-------------+---------------
estimated       200 MiB |    45 MiB   |    245 MiB  |
```

28. wget

Description: Downloads files from web servers (HTTP, HTTPS, FTP), supporting resuming interrupted downloads and quiet mode.

Syntax

```
wget [options] URL
wget URL
wget -O file URL
wget -c URL
wget -q URL
```

Terminal

```
isha@Ubuntu:~/Desktop$ wget -O linux-guide.pdf https://example.com/files/linux-guide.pdf
--2026-09-11 09:32:10--  https://example.com/files/linux-guide.pdf
Resolving example.com (example.com)... 93.184.215.14
Connecting to example.com (example.com)|93.184.215.14|:443... connected.
HTTP request sent, awaiting response... 200 OK
Length: 2456012 (2.3M) [application/pdf]
Saving to: 'linux-guide.pdf'

linux-guide.pdf     100%[===================>]   2.34M  4.21MB/s    in 0.6s

2026-09-11 09:32:11 (4.21 MB/s) - 'linux-guide.pdf' saved [2456012/2456012]
```

Summary Table — All Commands

No.	Command	Purpose	Important Options / Examples
1	arp	Displays and manages the system's ARP (Address Resolution Protocol) table, which maps IP addresses to MAC addresses on the local network.	arp -a | arp -n | arp -d <IP>
2	curl	Transfers data to or from a server; commonly used to test URLs, download files, and interact with REST APIs.	curl URL | curl -I URL | curl -v URL | curl -O URL | curl -X GET/POST URL
3	host	Performs DNS lookups to resolve domain names to IP addresses, or to query specific DNS record types.	host domain.com | host -t A domain.com | host -t MX domain.com
4	hostid	Displays the unique 32-bit hexadecimal identifier assigned to the host.	hostid
5	hostname	Displays or sets the system's hostname; can also show IP addresses associated with the host.	hostname | hostname -I | hostname -i | hostname -f
6	hostnamectl	Views or changes the system hostname and related settings, managed by systemd.	hostnamectl | hostnamectl status | hostnamectl set-hostname NAME
7	ifconfig	Legacy tool to view or configure network interfaces (IP address, netmask, status).	ifconfig | ifconfig eth0 | ifconfig -a | ifconfig eth0 up/down
8	iftop	Displays a live, continuously updating view of bandwidth usage on a network interface, per connection.	sudo iftop | iftop -i eth0 | iftop -n | iftop -P
9	ip	Modern replacement for ifconfig/route; manages IP addresses, links, routing tables, and neighbor (ARP) entries.	ip a | ip link | ip route | ip neigh
10	ipcrm	Removes System V inter-process communication (IPC) resources such as shared memory, message queues, or semaphores.	ipcrm -m ID | ipcrm -q ID | ipcrm -s ID
11	ipcs	Displays information about active System V IPC resources: shared memory segments, message queues, and semaphores.	ipcs | ipcs -m | ipcs -q | ipcs -s
12	iptables	Configures the Linux kernel firewall — sets rules to allow, block, or forward network packets.	iptables -L | iptables -L -n -v | iptables -S | iptables -t nat -L
13	iptables-save	Dumps the current iptables rule set to standard output in a format that can be restored later.	iptables-save | sudo iptables-save > rules.txt
14	iwconfig	Views or configures wireless network interface parameters such as SSID, mode, and signal strength.	iwconfig | iwconfig wlan0 | iwconfig wlan0 essid NAME
15	nc (netcat)	Reads and writes raw data across network connections; used to test open ports, transfer data, or set up simple listeners.	nc -zv HOST PORT | nc -l PORT | nc -u HOST PORT

No.	Command	Purpose	Important Options / Examples
16	netstat	Displays active network connections, listening ports, and network statistics (legacy tool, superseded by ss).	netstat -a | netstat -t | netstat -u | netstat -l | netstat -tulpn
17	nmcli	Command-line interface for NetworkManager; manages network devices, connections, and Wi-Fi.	nmcli device status | nmcli connection show | nmcli device wifi list
18	nslookup	Queries DNS servers to resolve domain names to IP addresses or vice versa.	nslookup domain.com | nslookup domain.com 8.8.8.8
19	ping	Tests connectivity to a host by sending ICMP echo requests and measuring the response time.	ping HOST | ping -c 4 HOST | ping -4 HOST | ping -6 HOST
20	rcp	Copies files between remote systems using the older, unencrypted 'r' protocol (largely replaced by scp/rsync).	rcp file user@host:/path | rcp -r folder user@host:/path
21	route	Displays or manually manipulates the IP routing table (legacy tool, superseded by ip route).	route -n | route | route add | route del
22	rsync	Efficiently synchronizes or copies files and directories, locally or over a network, transferring only changed data.	rsync -av | rsync -avn | rsync -e ssh
23	scp	Securely copies files between hosts over an encrypted SSH connection.	scp file user@host:/path | scp -r | scp -P PORT
24	ssh	Establishes a secure, encrypted remote login session or executes commands on a remote host.	ssh user@host | ssh -p PORT | ssh -i KEY | ssh -v
25	tracepath	Traces the network path to a destination host and reports MTU along the route, without requiring root privileges.	tracepath domain.com | tracepath -n HOST
26	traceroute	Displays the route and measures transit delays of packets across a network to a destination host.	traceroute HOST | traceroute -n HOST | traceroute -I HOST
27	vnstat	Monitors and logs network traffic over time, providing daily, monthly, and historical usage statistics.	vnstat | vnstat -i eth0 | vnstat -d | vnstat -m
28	wget	Downloads files from web servers (HTTP, HTTPS, FTP), supporting resuming interrupted downloads and quiet mode.	wget URL | wget -O file URL | wget -c URL | wget -q URL